Mehedi Hasan

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EDUCATION

Jahangirnagar University *Aug 2023 – Sep 2024*
MSc. in Computer Science & Engineering ***GPA: 3.50/4.0***

Jahangirnagar University *Feb 2018 – July 2023*
BSc. in Computer Science & Engineering ***GPA: 3.47/4.0***

EXPERIENCE

Samsung R&D Institute Bangladesh *Dhaka, Bangladesh*
Senior Software Engineer *Jan 2024 – Present*

Research Experience | *Key Fields: ML, NLP, Agentic AI, Wearable Devices*

- **Agentic Prior-Art Search and Patent-Idea Similarity Reasoning**
  - Developed an interactive AI agent for prior-art search to retrieve relevant patents and assess the novelty of proposed ideas.
  - Designed a generator–validator framework with an adaptive knowledge graph to iteratively refine ideas and identify potential obviousness relationships across multiple patents.
  - Developed a structure-aware RAG pipeline utilizing patent document sections and weighted semantic relationships among technical claims, methodologies, and user-proposed ideas to support evidence-grounded similarity reasoning.
- **Wearable-Based Liquid Consumption Estimation Using Machine Learning**
  - Developed a ML solution for estimating liquid intake from wearable sensor (accelerometer and gyroscope) data.
  - Developed a data augmentation approach combining gulp patterns to address limited training data and improve generalization to unseen individuals.
  - Investigated normalization techniques for accelerometer and gyroscope signals across different wearable devices, analyzing the impact of sensor variability on estimation performance.
- **Context-Aware On-Device Widget Positioning for Galaxy Watch**
  - Developed a context-aware solution for dynamically positioning floating widgets on the Samsung Galaxy Watch watchface, minimizing information loss and maximizing visual balance.
  - Fine-tuned an on-device machine-learning model to classify underlying watchface widgets and guide adaptive positioning.

Development Experience | *Tech Stack: Android, WearOS, RTOS*

- Contributed to Samsung Galaxy Watch 7/8/9 releases through requirements analysis, feasibility studies, prototyping, and feature development; mentored junior engineers and resolved 300+ SQE- and market-reported issues.
- Developed core Galaxy Watch modules including notifications, watchface overlay, backup-and-restore, and settings; migrated XML interfaces to Jetpack Compose, integrated WearOS with Samsung's low-power AOD hardware, and designed the ongoing-activity management backend architecture.

PUBLICATIONS

*Equal contribution

- *M. Hasan, M. I. Mahim, J. K. Das (2026).* **Segmentation Strategies for Long-Code Authorship Detection: Balancing Context Coverage and Computational Efficiency.** *Jahangirnagar University Journal of Electronics and Computer Science.* **| Paper | Code**
  - Compared truncation, multi-view cropping, and sliding-window strategies for transformer-based long-code authorship detection, analyzing context coverage, classification performance, and computational efficiency.
- *M. I. Mahim*, M. Hasan* (2025).* **TeamHateMate at BLP Task1: Divide and Conquer: A Two-Stage Cascaded Framework with K-Fold Ensembling for Multi-Label Bangla Hate Speech Classification.** *Proceedings of the 2nd Workshop on Bangla Language Processing (BLP), AACL-IJCNLP 2025.* **| Paper | Code | Best Shared Task Paper Award**
  - Proposed a two-stage cascaded framework with k-fold ensembling to address class imbalance in low-resource Bangla hate-speech classification across hate-speech type, target, and severity dimensions, achieving **0.7392 micro-F1**.

SELECTED PROJECTS

- **Sleep-Onset Prediction from Wearable EEG**
  - Developed causal multi-scale CNN–GRU models to estimate seconds until N2 sleep onset from Muse headband EEG, reaching a public-test weighted binned MAE of 47.74 on Neural Interfaces 2026 Track 3.
- **Human–Machine–Hybrid–Adversarial Code Classification**
  - Developed an adaptive training loop prioritizing repeatedly misclassified samples to address class imbalance and investigate robustness across human-written, AI-generated, hybrid, and adversarial code.
- **Layerwise Probing and Ablation Analysis in BERT**
  - Investigated BERT's 12 encoder layers through linear probing and ablation on POS tagging and semantic similarity, finding that middle layers yield more separable representations while upper layers influence predictions.
- **Machine Learning-Based Image Forgery Detection**
  - Developed an image-forgery detection pipeline combining LBP-DCT features with an ensemble of SVM, Random Forest, and Gradient Boosting, outperforming individual classifiers.

COMPETITIONS & ACHIEVEMENTS

Competitive Programming & Problem Solving

- Solved **1,900+ algorithmic problems** across multiple online judges, covering mathematics (484), greedy algorithms (544), dynamic programming (126), data structures (110), graph algorithms (67), and number theory (87).
- Participated in **160+ programming contests**; achieved a peak rating of **1611 (Expert)** on Codeforces (`@MhD88`) and **1730 (3-star)** on CodeChef (`@mhd88`).
- Ranked in the **top 2%** among 12,275 participants and 16,010 participants in Codeforces contests.

Hackathons & ML Contests

- Became a **finalist among 400+ teams** at the Code Samurai Hackathon 2022, hosted by the University of Dhaka; developed a web-based system in the final round.
- Ranked **1st among 90 teams** (Team: `team_hate_mate`) at the BLP Workshop 2025 Shared Task.

SCHOLARSHIPS & AWARDS

- Received **two Samsung Excellence Awards** (2024, 2025) for innovation and development, and earned **Professional-Level SWC Certification** (2024).
- Awarded **Dean's Merit Scholarships** throughout all four years of B.Sc. study (2018–2023).
- Received **National Merit Scholarships** in primary and secondary school examinations (2011, 2013).

TEACHING EXPERIENCE

Competitive Programming Instructor | *Department of CSE, Jahangirnagar University* *2022 – 2023*

- Taught C/C++, data structures, algorithms, and problem-solving; developed practice materials and organized weekly contests.

Private Academic Tutor | *In-Home Tutoring* *2020 – 2024*

- Provided one-on-one tutoring in C/C++, OOP, data structures, and mathematics.

Online ICT Instructor | *Remote* *Jun 2020 – Aug 2020*

- Taught Information & Communication Technologies (programming fundamentals, number systems, database)

SKILLS

Programming Languages	: C, C++, Java, Kotlin, Python, JavaScript
ML & AI	: scikit-learn, NumPy, OpenCV, PyTorch, Hugging Face Transformers, LLMs
Frameworks & Technologies	: Android, Node.js, Jetpack Compose, HILT, Bootstrap
Databases	: MySQL, MongoDB
Tools	: Git, Perforce, VS Code, Android Studio